

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

SUBJECT : Cyber & Information Security

Aim: Web Security with SSL/TLS

Step 1: Open Cisco Packet Tracer Open Cisco Packet Tracer. You will see the workspace where you can add networking devices.

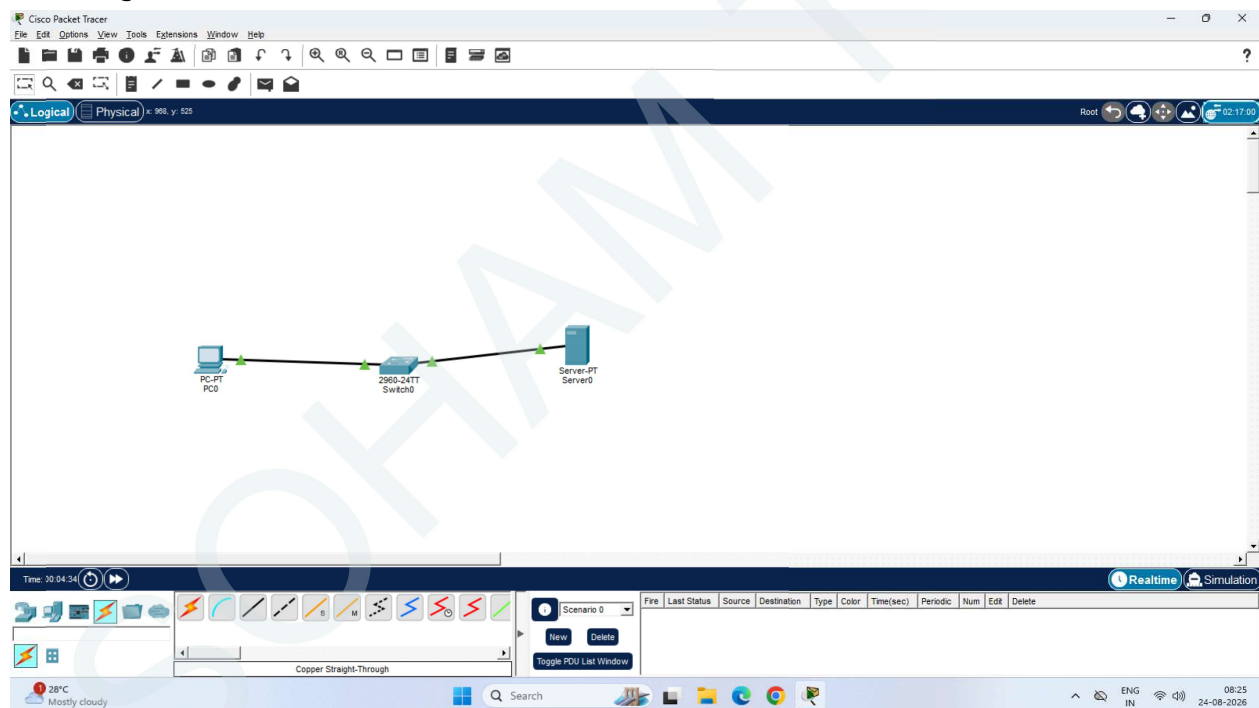
Step 2: Add a PC From the bottom-left device selection area: End Devices → PC Drag one PC into the workspace. The device will appear as: PC0

Step 3: Add a Server Go to: End Devices → Server Drag one server into the workspace. It will appear as: Server0

Step 4: Add a Switch Go to: Network Devices → Switches Select: 2960 Switch Drag it into the workspace. It will appear as: Switch0

Step 5: Connect the Devices Click the Connections icon. Select: Copper Straight-Through Connect PC to Switch PC0 FastEthernet0 | ▼ Switch0 FastEthernet0/1 Connect Server to Switch Server0 FastEthernet0 | ▼ Switch0 FastEthernet0/2

The final topology should be: PC0 | | Switch0 | | Server0 Wait until the connection indicators become green



Step 6: Configure IP Address on PC

Click:

PC0 → Desktop → IP Configuration

Enter:

Setting | Value

Soham Patil
Tycs T103

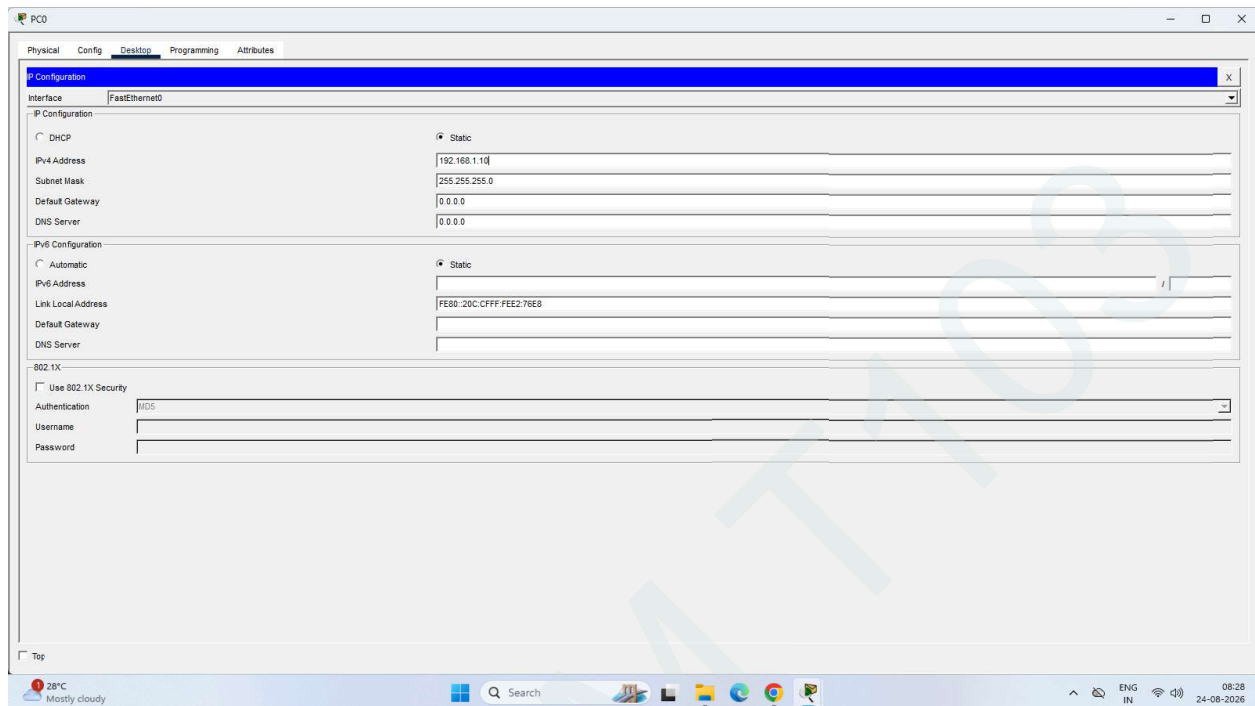
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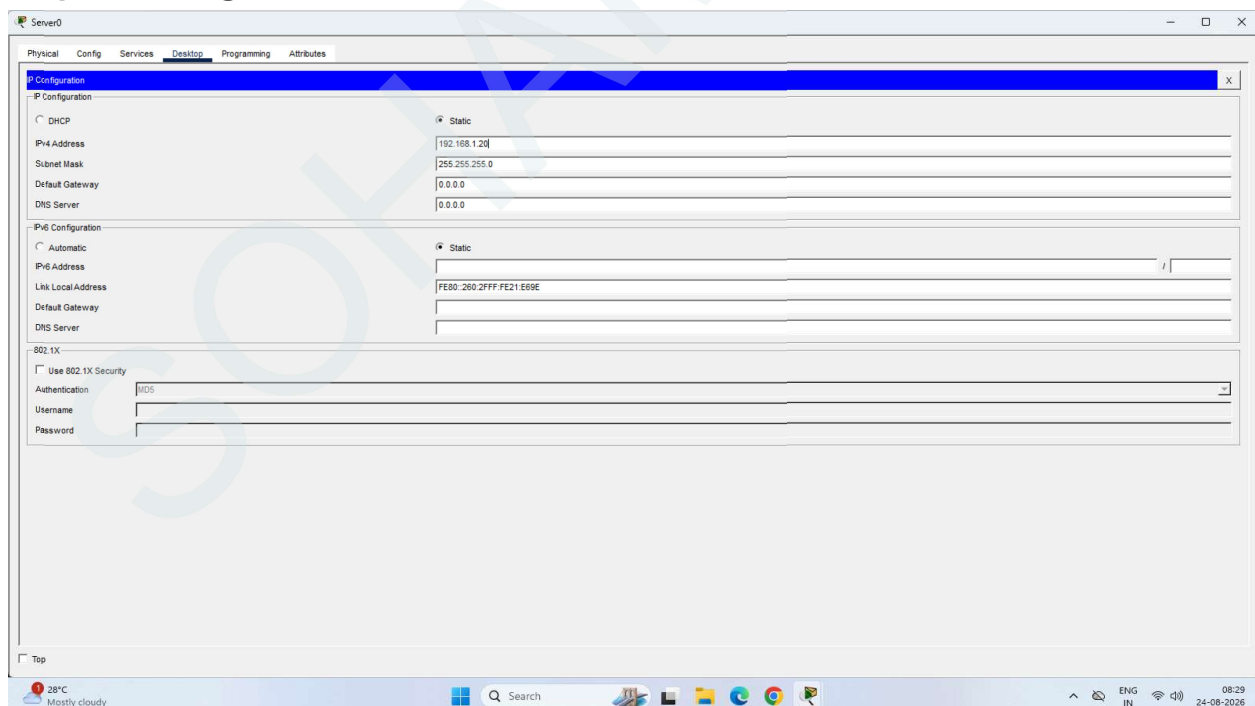
IP Address | 192.168.1.10

Subnet Mask | 255.255.255.0

Default Gateway | Leave blank



Step 7: Configure IP Address on Server

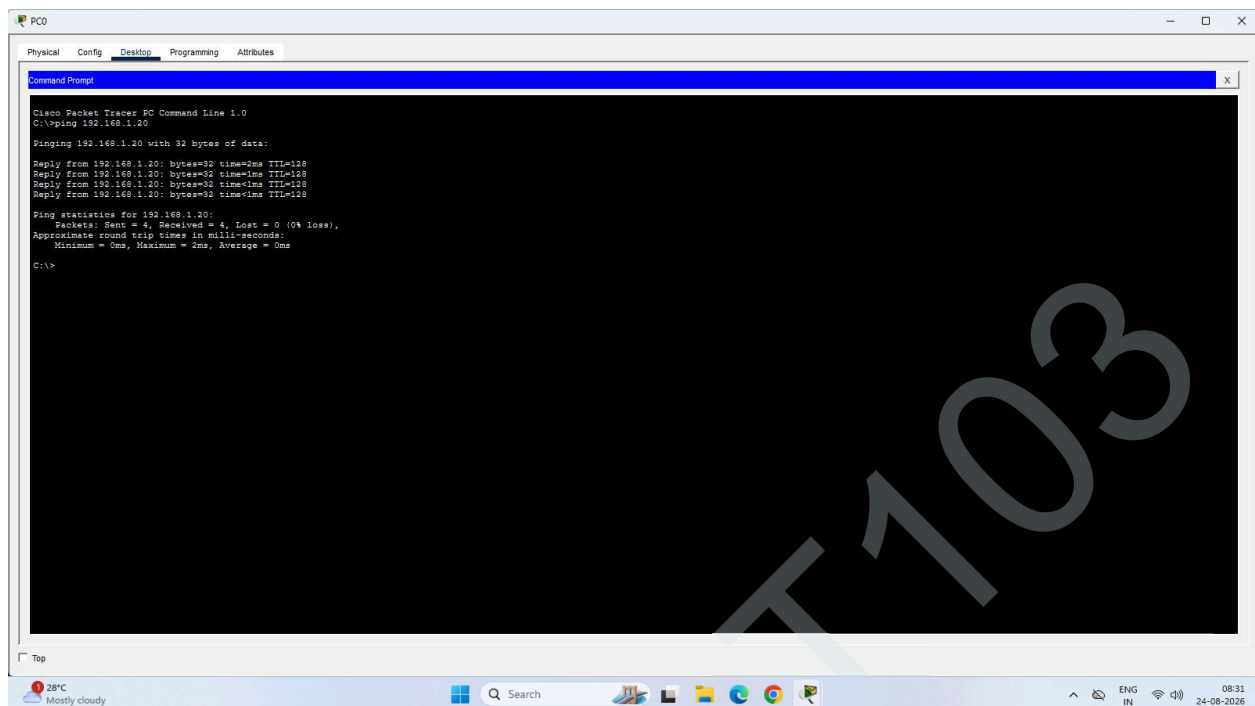


Step 8: Test Network Connectivity

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Step 9: Configure HTTP and HTTPS Service

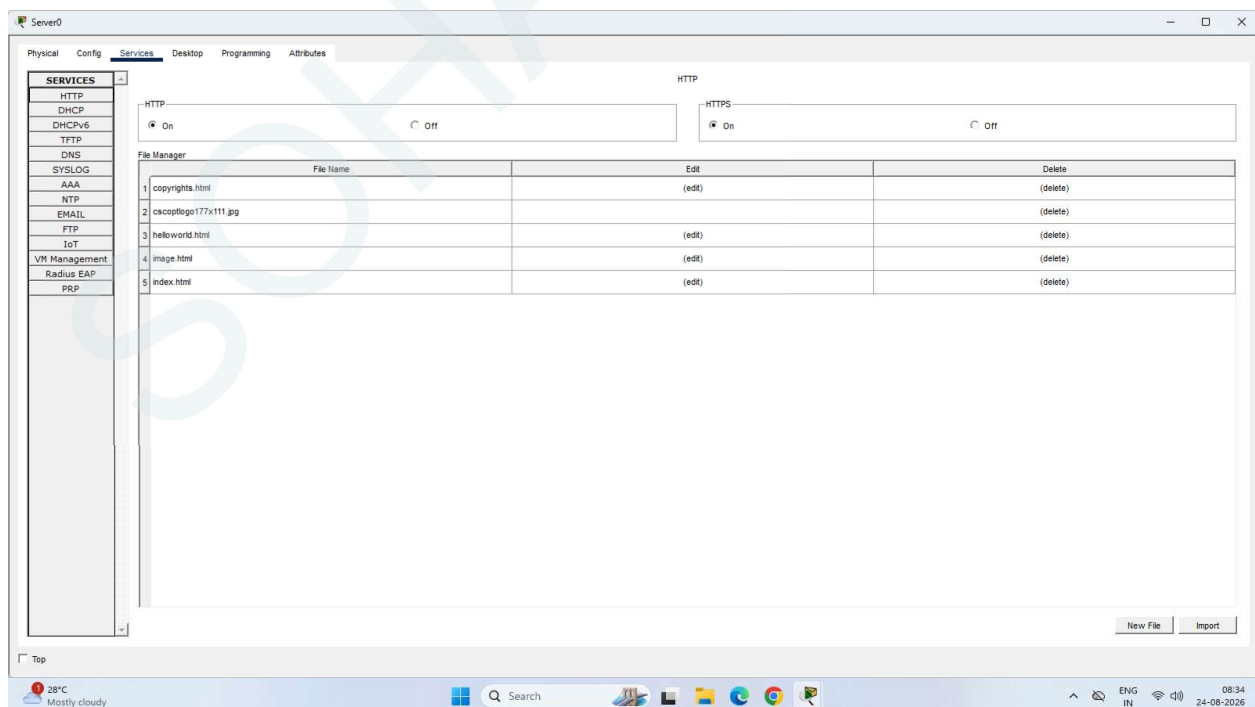
Click: Server0 → Services → HTTP You will see the HTTP service configuration.

Enable HTTP Set: HTTP: ON Enable HTTPS

Set: HTTPS: ON Now the server supports both:

HTTP → Port 80

HTTPS → Port 443



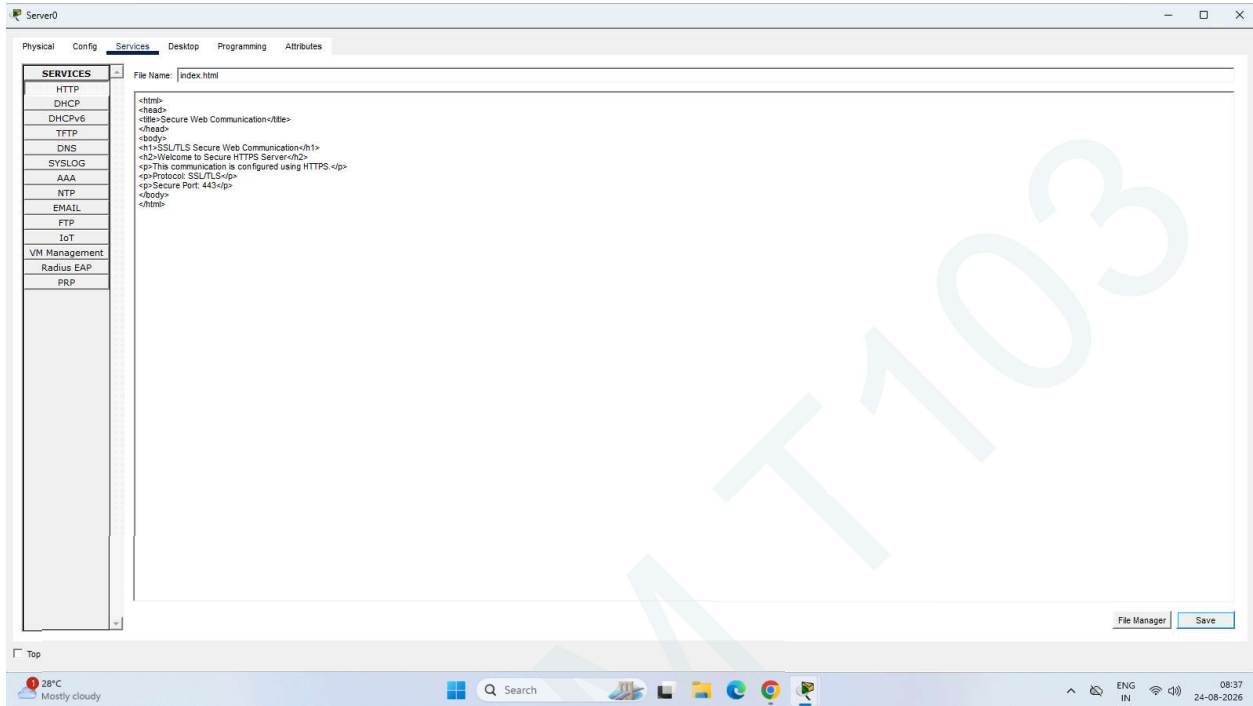
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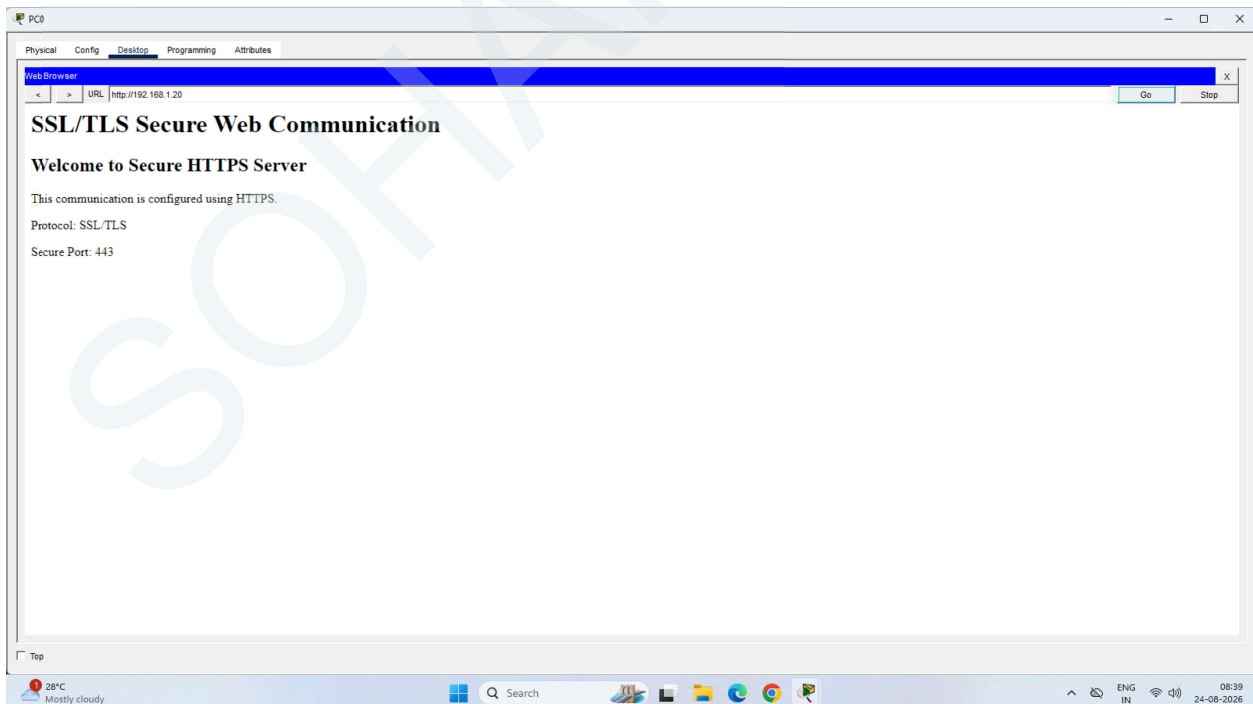
Step 10: Configure a Web Page

In: Server0 → Services → HTTP

You can edit the HTML web page



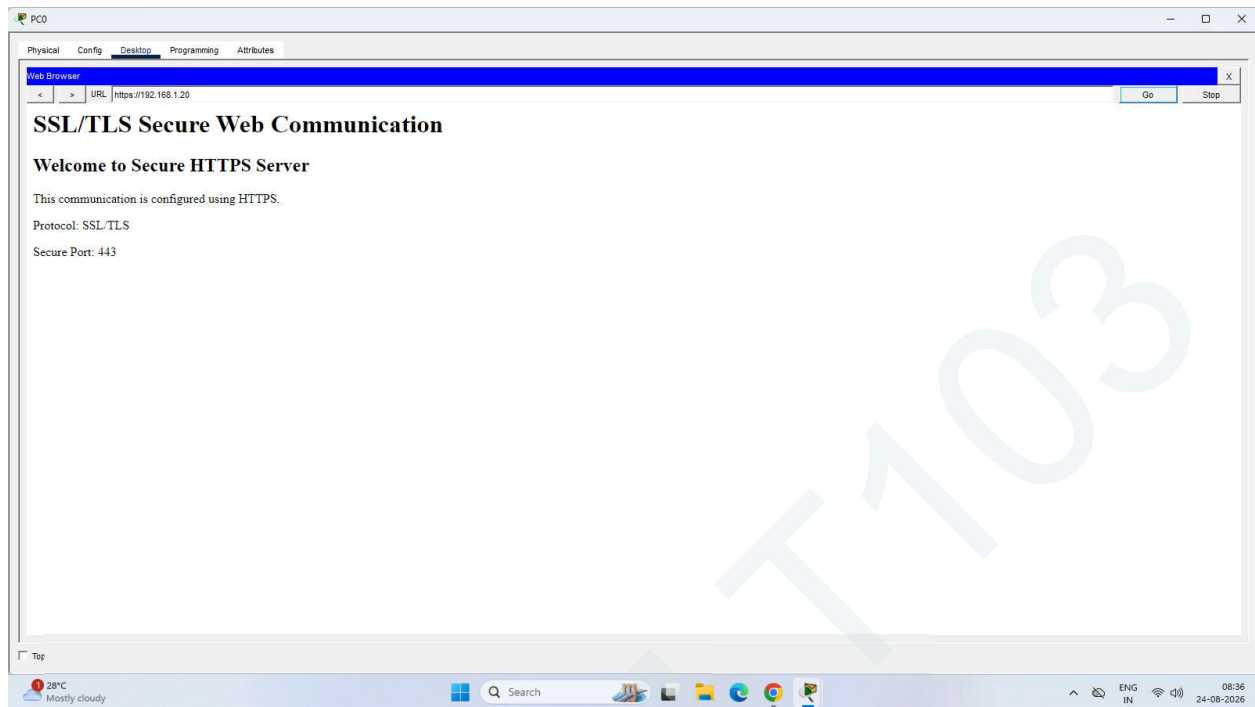
Step 11: Test Normal HTTP Communication



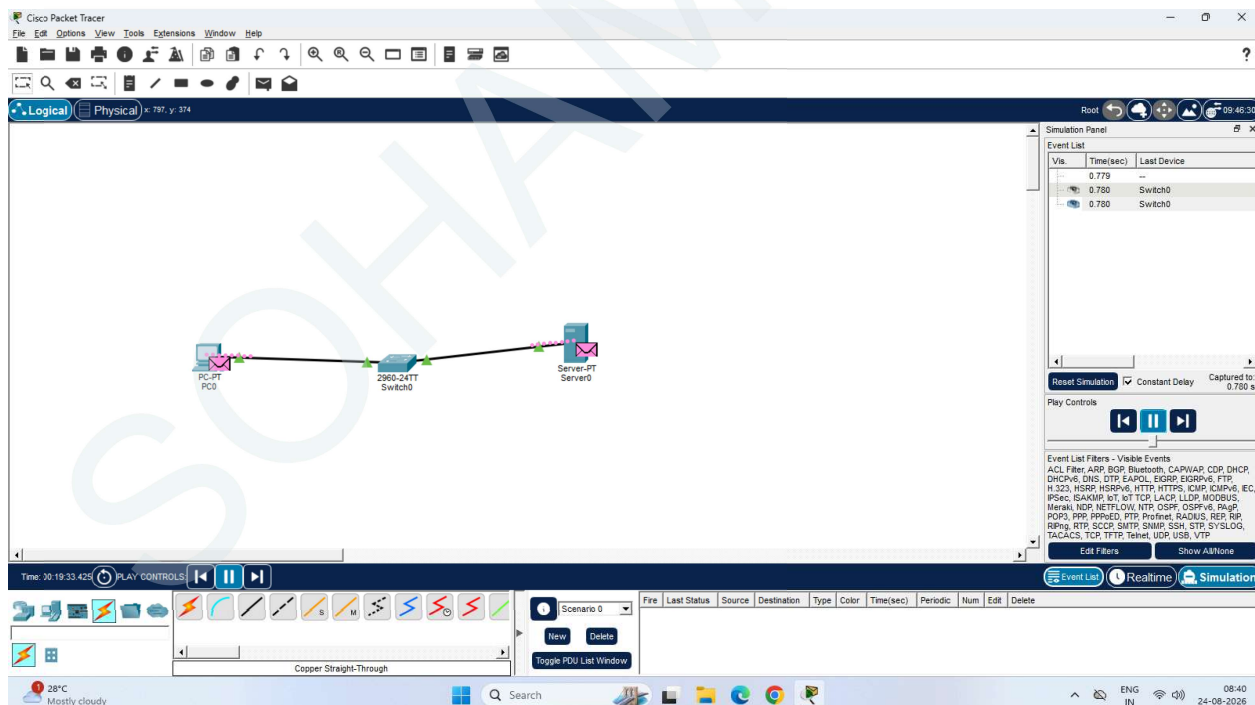
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Step 12: Test Secure HTTPS Communication



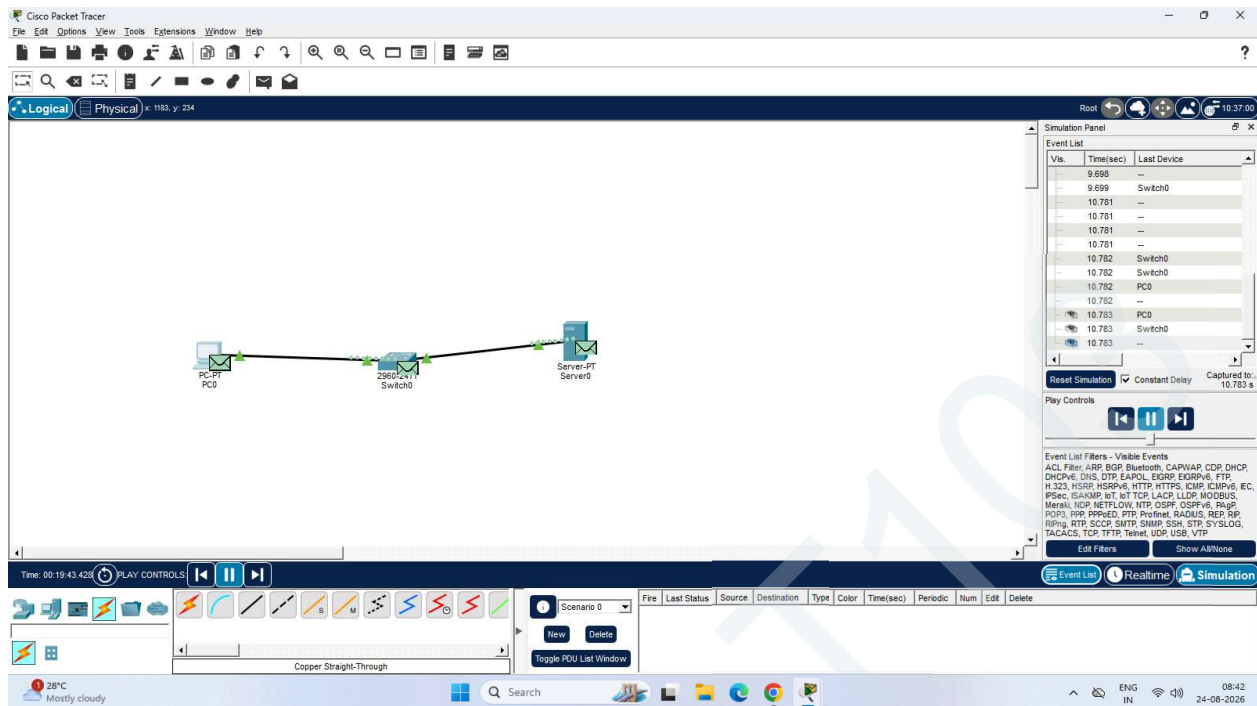
Step 13: Use Simulation Mode



Step 14: Observe HTTP Communication

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Step 15: Observe HTTPS Communication

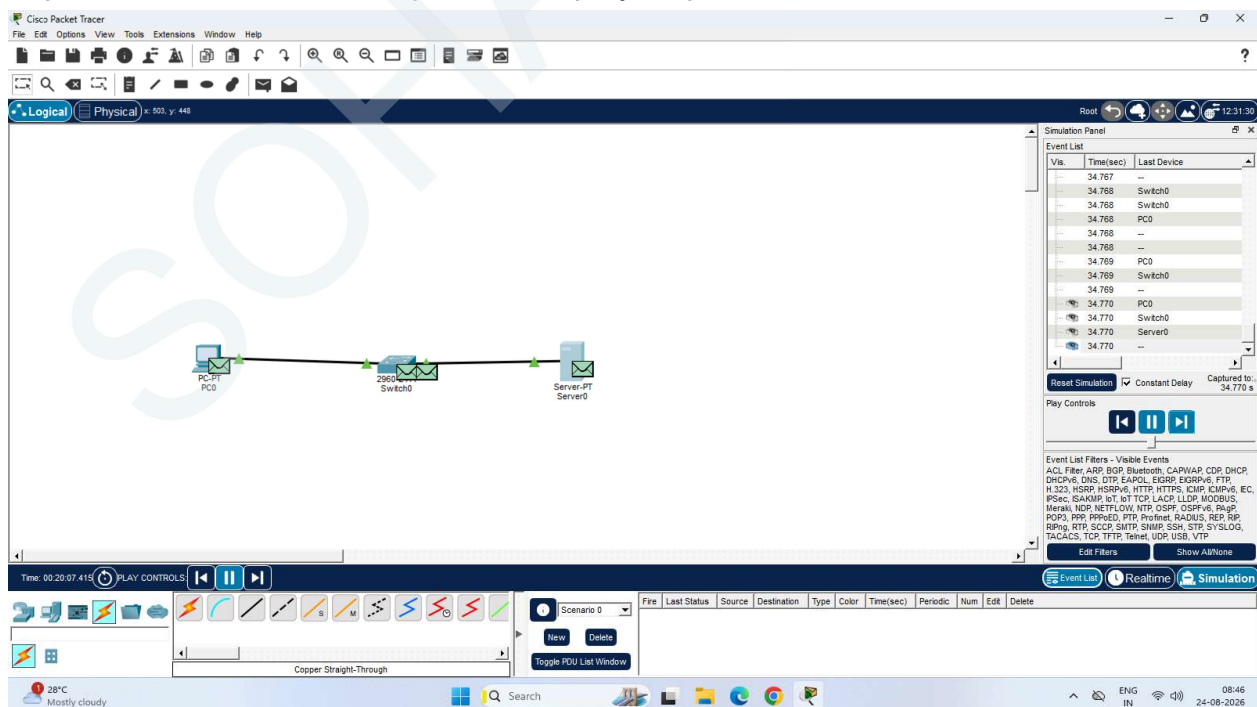
Now enter: `https://192.168.1.20`

The communication occurs between: PC0 | ▼ Switch0 | ▼ Server0

The server provides secure web service through: HTTPS Port 443

You can use:

Capture/Forward to observe packets step by step.



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The client PC successfully communicates with the web server using:

https://192.168.1.20

The HTTPS service is enabled on the server, demonstrating secure web communication using the SSL/TLS concept and port 443.

Result

Secure web communication was successfully configured and implemented in Cisco Packet Tracer by enabling HTTPS on the server and accessing the web server securely from the client PC. HTTP and HTTPS communication were tested using ports 80 and 443 respectively. The SSL/TLS secure session establishment and certificate management concepts were also studied